

```
resource "azurerm_virtual_network" "core" {  
  name = "vnet-prod-01"  
  address_space = ["10.0.0.0/16"]  
  location = var.region  
}
```

```
module "aks" {  
  source = "../modules/compute/aks"  
  cluster_name = "aks-prod-01"  
  node_count = 5  
}
```

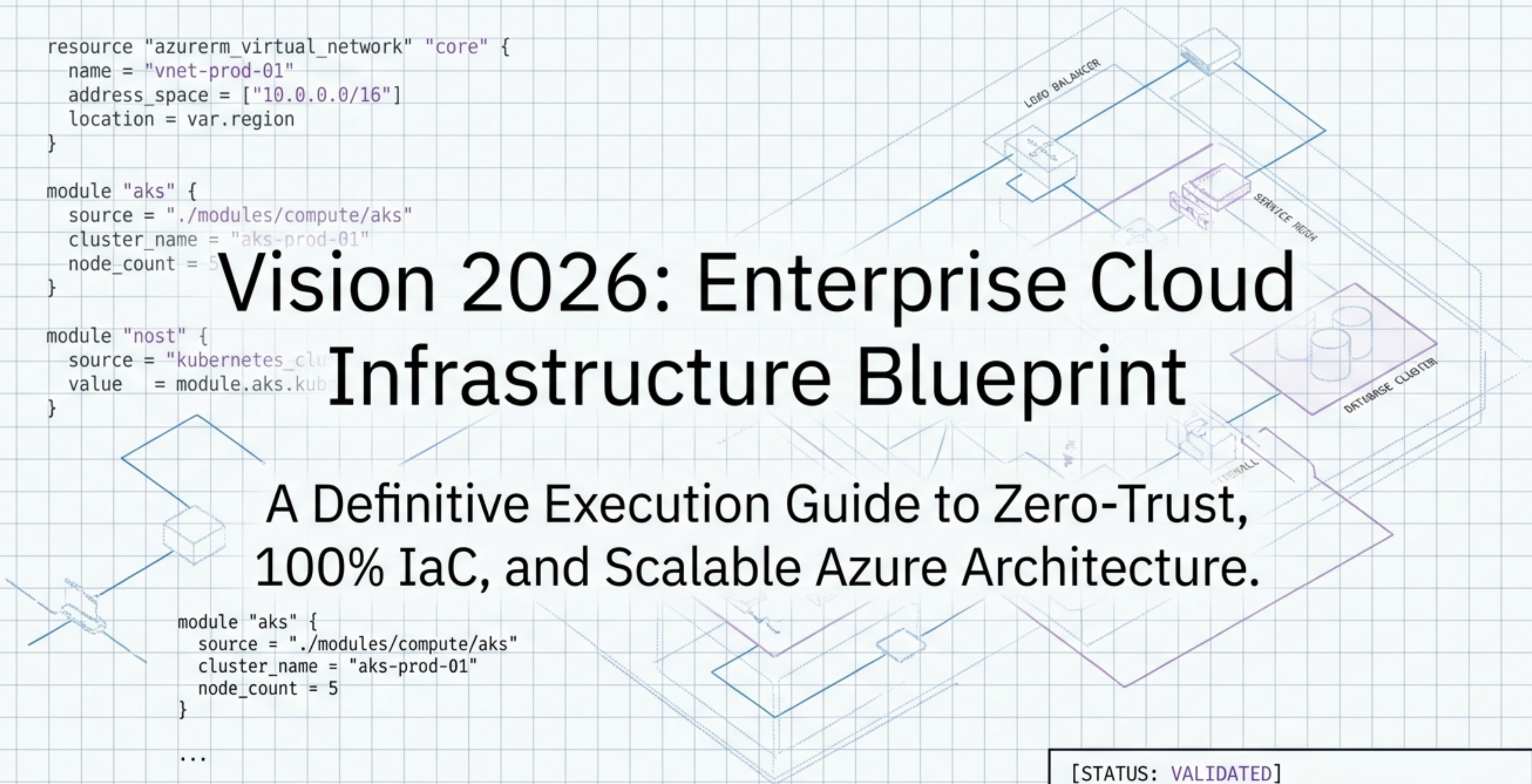
```
module "nost" {  
  source = "kubernetes_clu"  
  value = module.aks.kub
```

Vision 2026: Enterprise Cloud Infrastructure Blueprint

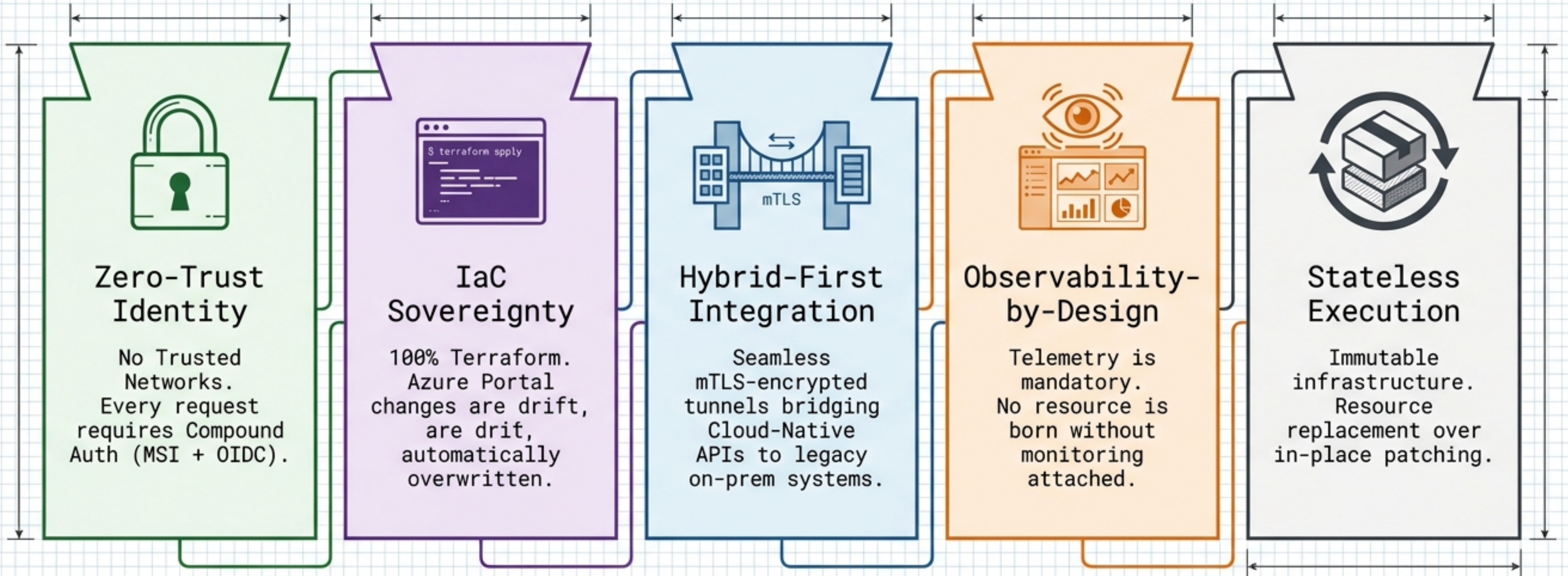
A Definitive Execution Guide to Zero-Trust,
100% IaC, and Scalable Azure Architecture.

```
module "aks" {  
  source = "../modules/compute/aks"  
  cluster_name = "aks-prod-01"  
  node_count = 5  
}  
  
...  
  
output "kubernetes_cluster_endpoint" {  
  value = module.aks.kube_config.0.host  
}
```

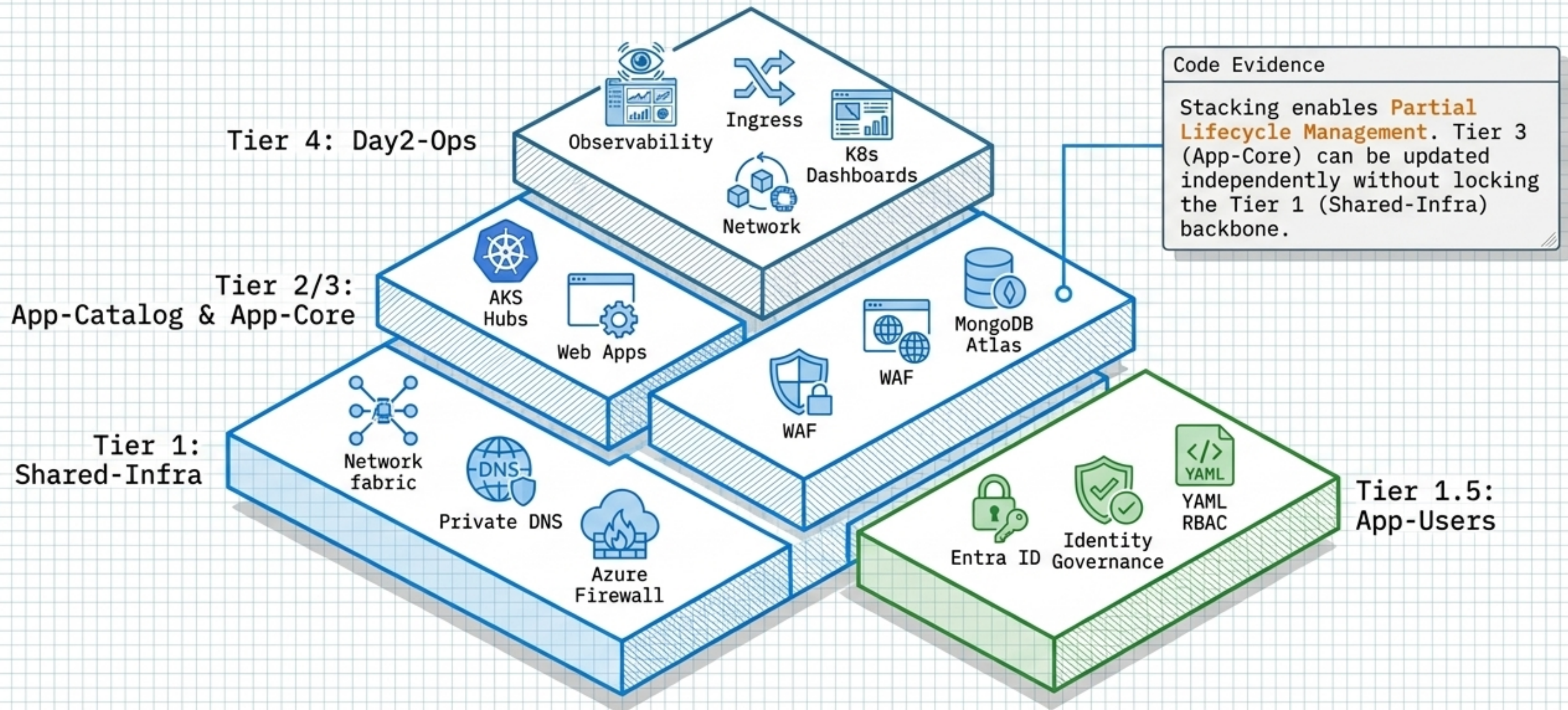
[STATUS: **VALIDATED**]
| repo: [nubenetes/terraform-azure-devops](#)
| version: 1.0.0



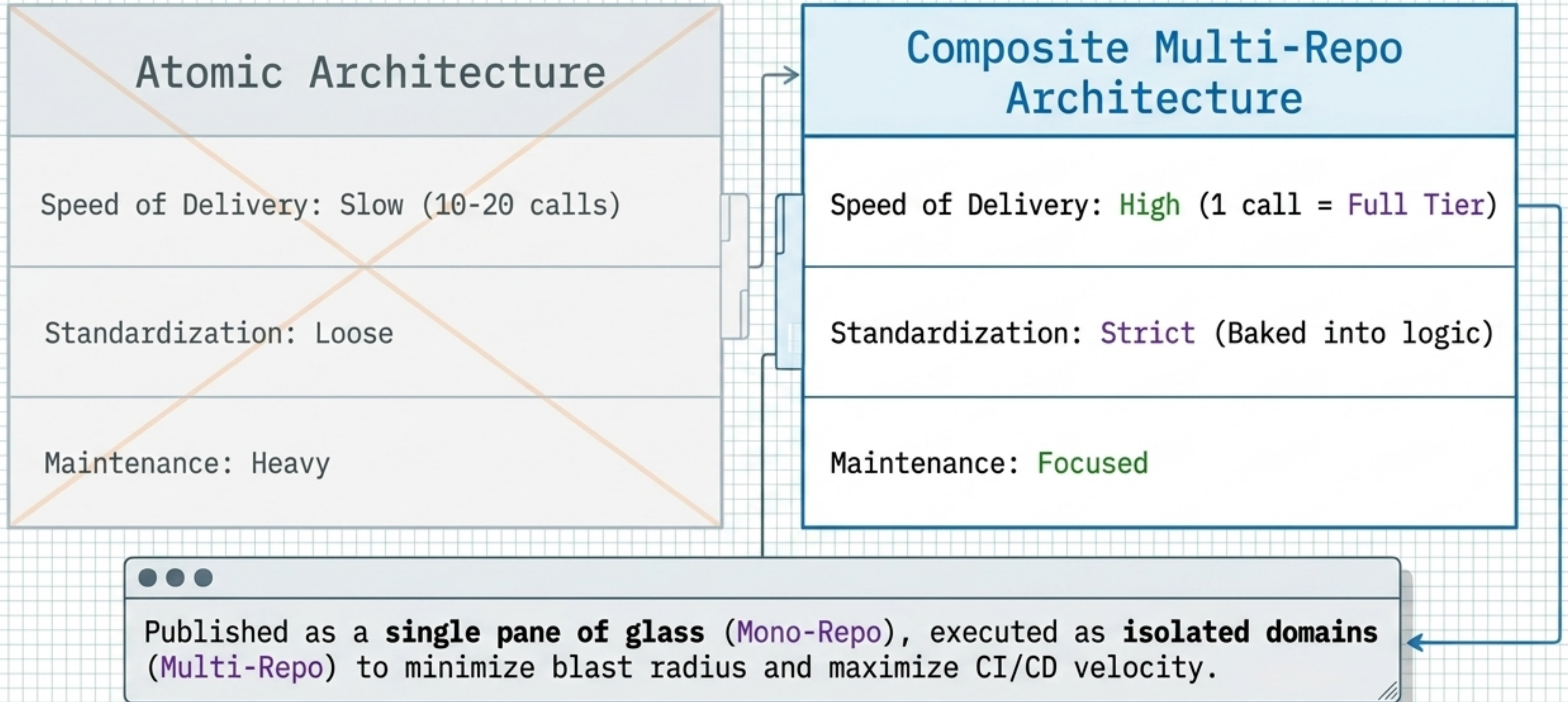
The Immutable Laws of Vision 2026



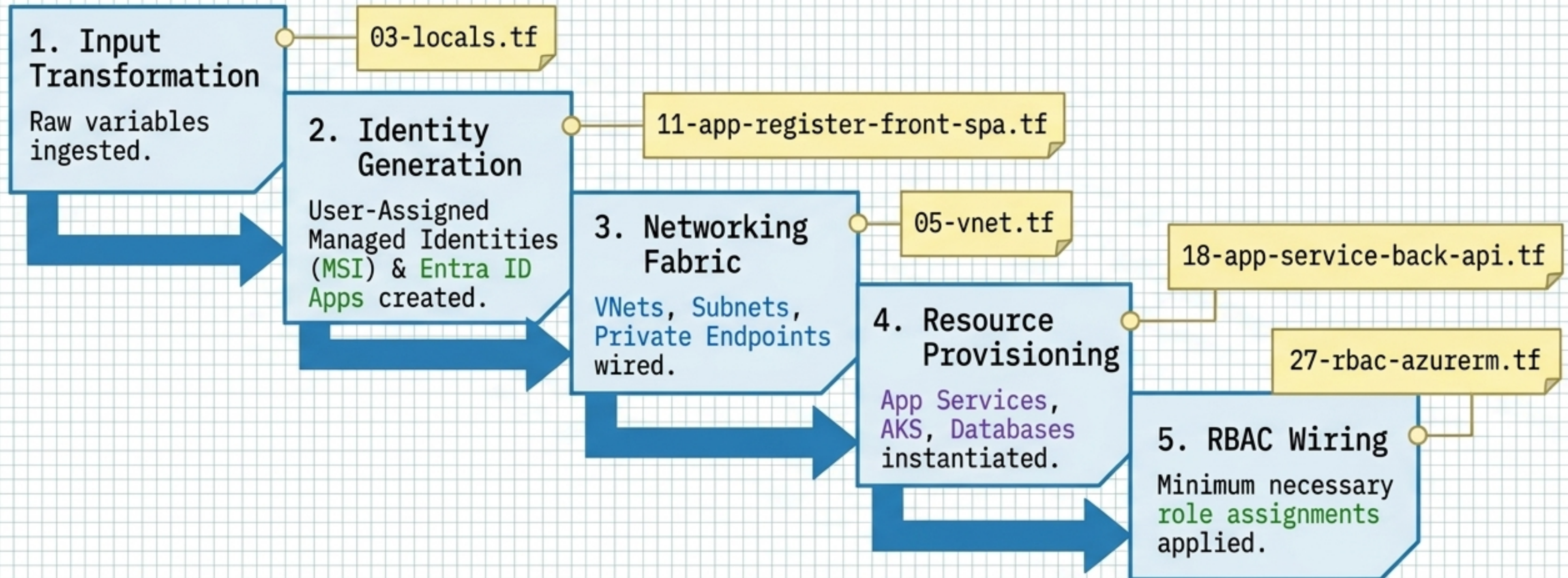
Architectural Dependency Hierarchy



Module Architecture: Domain-Driven Composition

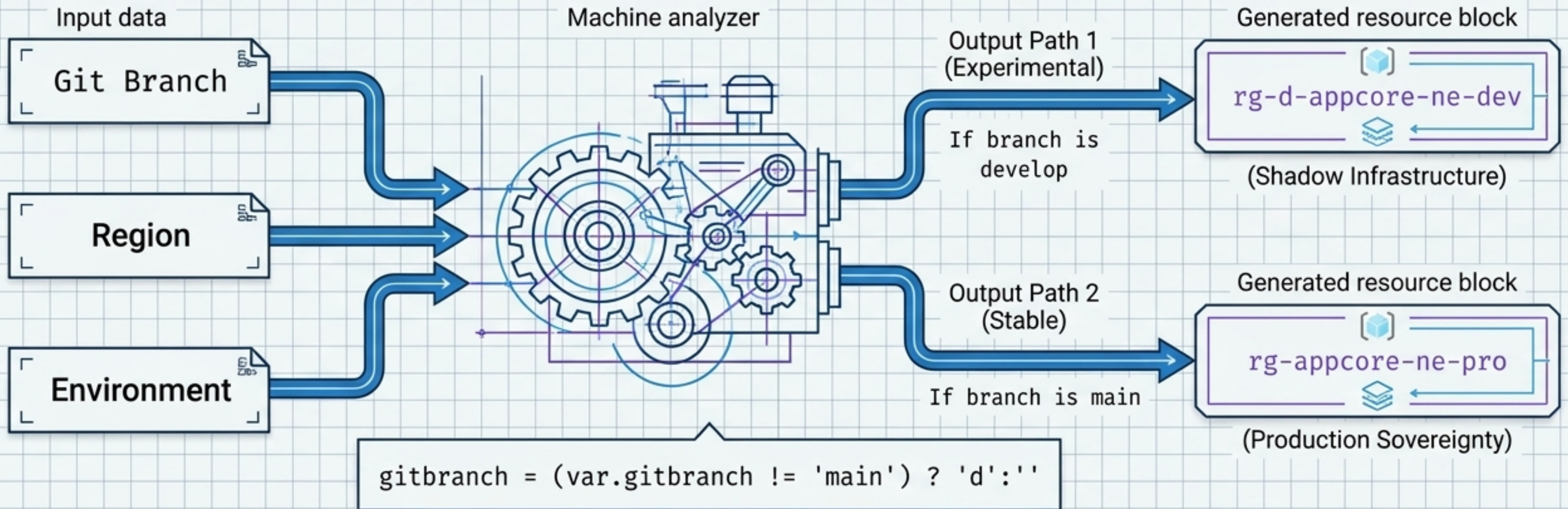


The 5-Stage Internal Execution Flow



Resources are never deployed until their security perimeter and identities are established.

The Logic Engine: IaC Sovereignty via the 'd' Prefix



Platform Engineers can safely test core IaC logic across all tiers **without ever risking collisions with production-grade resources.**

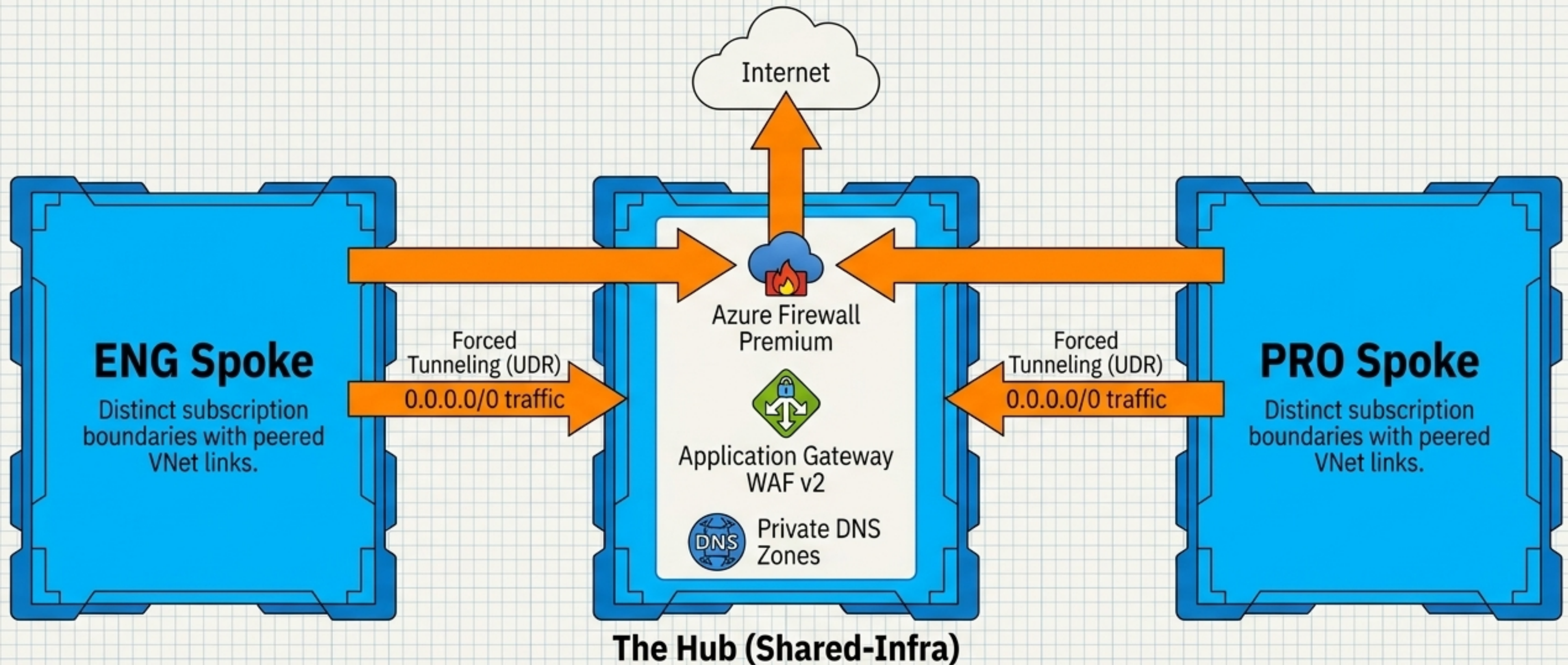
Tier Parity Strategy: ENG vs. PRO

Engineering (ENG)	 Control Switch	Production (PRO)
B1 / P1v2	App Service	P2v2 / P3v3
Standard_DS2_v2	AKS Node Pool	Standard_D4s_v3 (Auto-healing)
M10 (Standard)	MongoDB Tier	M30+ (High-Availability)
LRS (Local)	Replication	GRS / ZRS (Geo/Zone)
7 Days Retention	Backup	180+ Days Retention

Code Evidence Tag

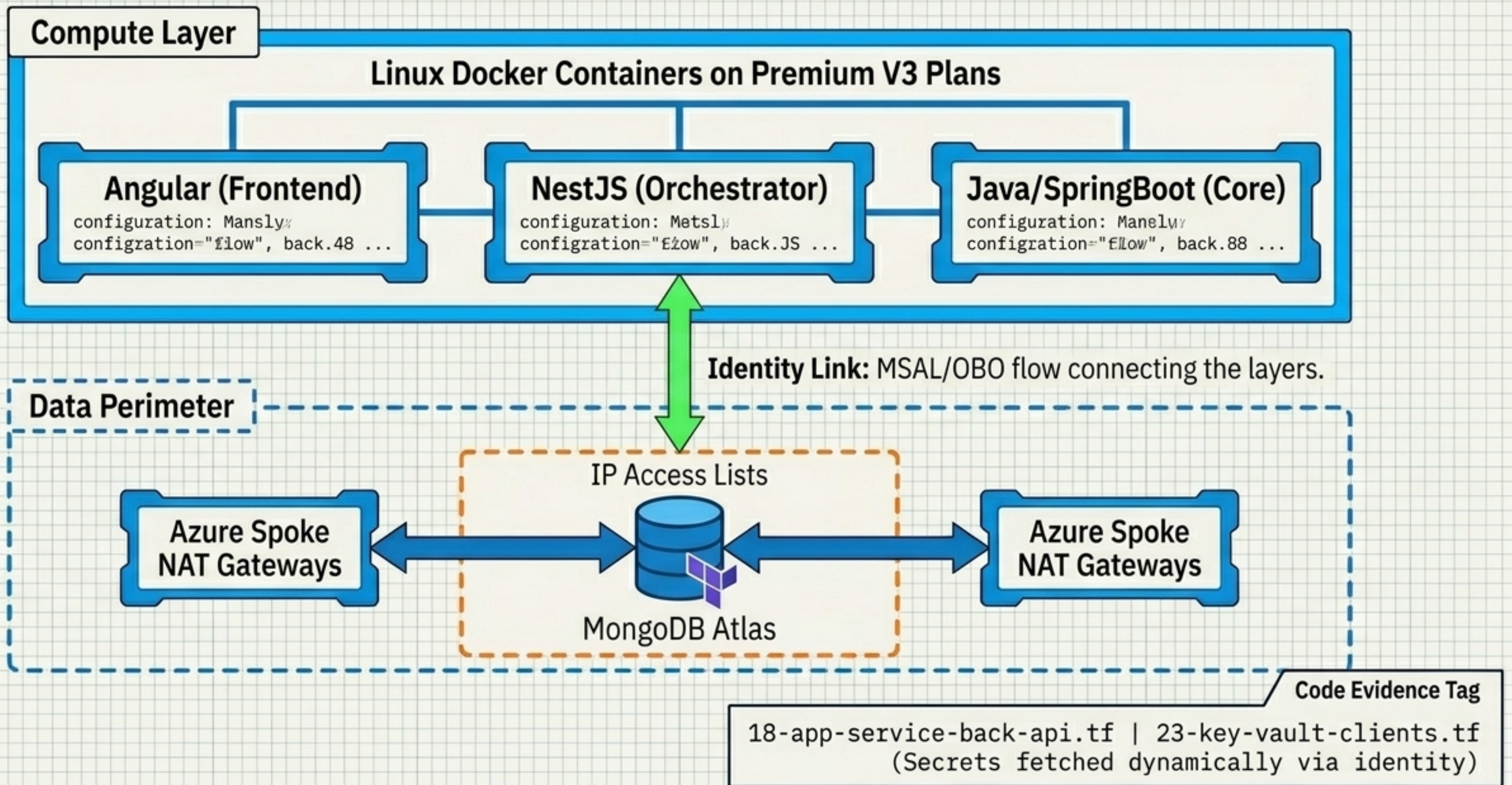
Variables bound to Enterprise Subscriptions: `svccon-ent-dev` vs `svccon-ent-pro`

The Symmetrical Hub-Spoke Backbone



Symmetrical hubs isolate environments for billing, but centralize egress for absolute Zero-Trust inspection.

Polyglot Microservices & Secure Persistence



AKS Machine Learning Inference Ecosystem

Network Isolation: snet-aks-* subnets

Compute Isolation: Dedicated appspool002 Node Pool

High-Memory/GPU Series
app=mljobs

1. Kubelet MSI pulls image from ACR via AcrPull RBAC.



ACR



2. Pod authenticates to Key Vault via Workload Identity.



Key Vault

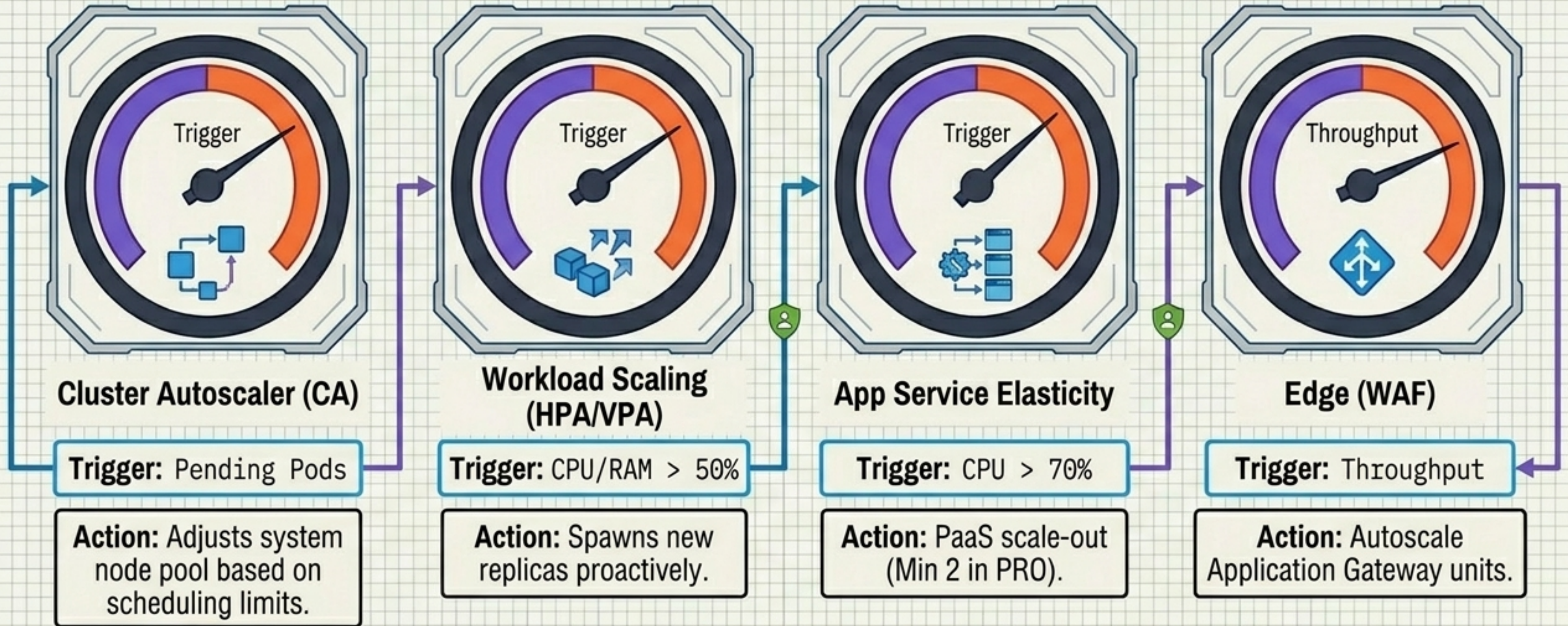
3. Pod downloads multi-GB model weights from Blob Storage via Private Link.



Blob Storage

AI assets and datasets are never exposed to the public internet during inference.

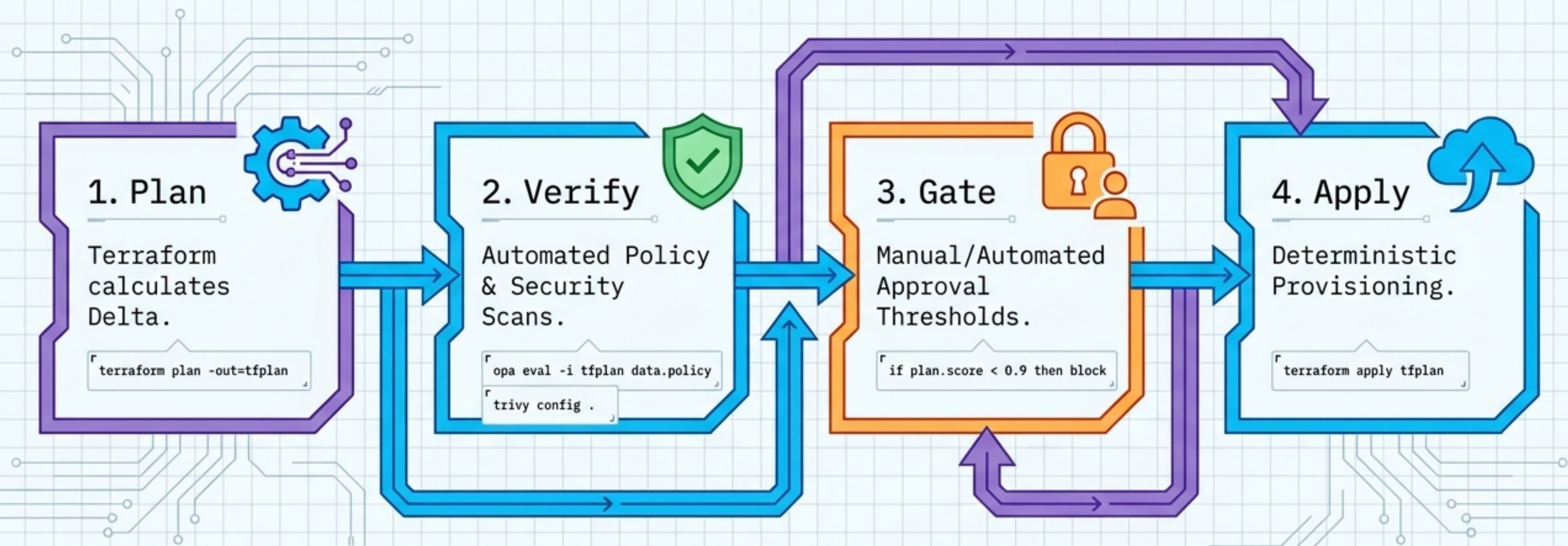
Four-Tier Scaling Intelligence



Code Evidence Tag

06-aks-cluster.tf (expander = 'most-pods')

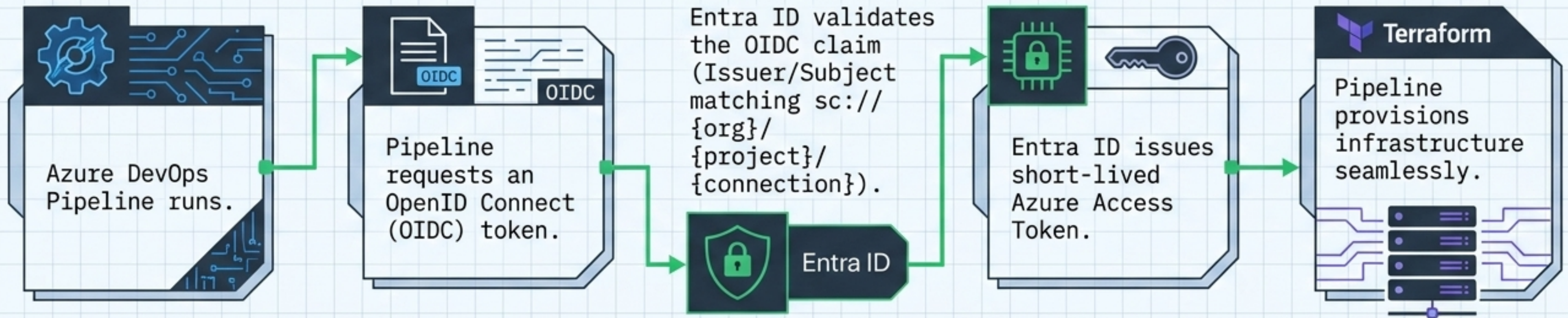
The Immutable Deployment Execution Flow



A strict **Plan-First, Verify-Always** execution model ensures production is never compromised by unverified changes.

Zero-Secret CI/CD Trust Fabric

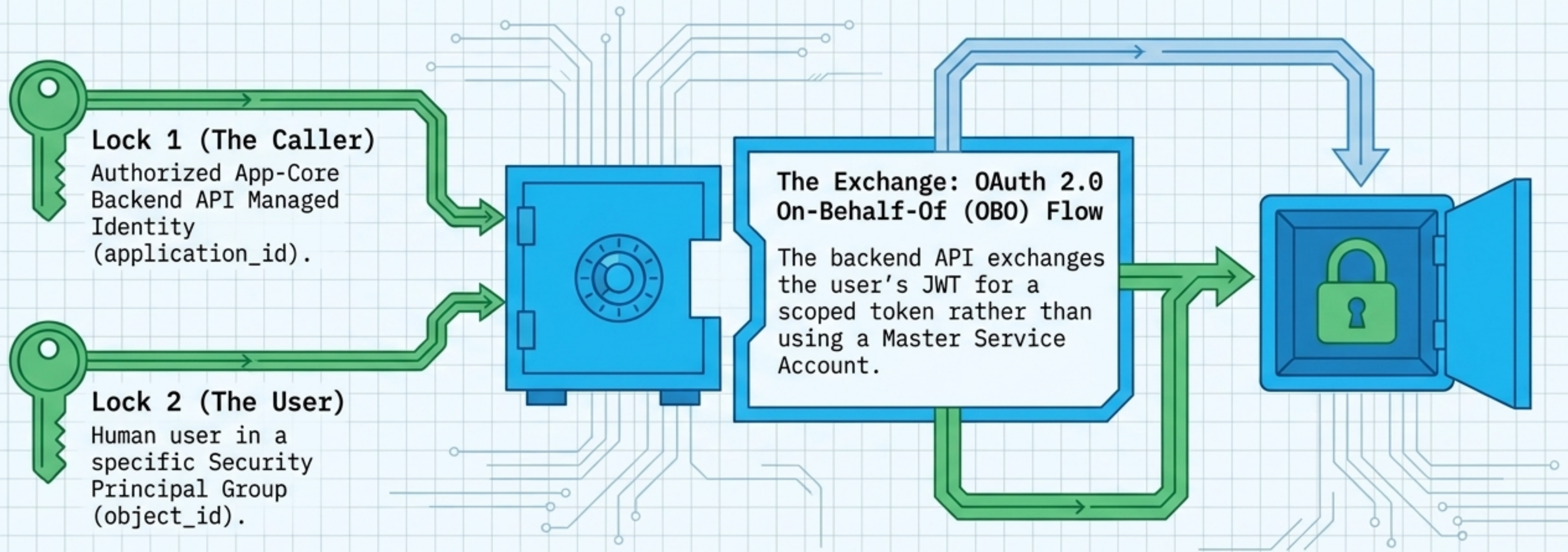
Workload Identity Trust Chain



Code Evidence Tag

```
use_oidc = true applied across all  
core providers.
```


Advanced DevSecOps: Compound Identity & OBO Flow



Insight: **Identity Theft** by a rogue application or **Direct Access** by an unauthorized user are mathematically prevented. Both conditions must be met.

Orchestration Evolution: Terraform 1.15 Dynamic Sources

Legacy

```
module "s3_bucket" {  
  source = "terraform-aws-modules/s3-bucket/aws"  
  version = "3.14.1"  
  # ... other parameters  
}
```

Requires duplicating the entire module block 3x across DEV/QA/PRO.

DEV

```
module "s3_bucket" {  
  source = "terraform-aws-modules/s3-bucket/aws"  
  version = "3.14.1"  
  # ... other parameters  
}
```

QA

```
module "s3_bucket" {  
  source = "terraform-aws-modules/s3-bucket/aws"  
  version = "3.14.1"  
  # ... other parameters  
}
```

PRO

```
module "s3_bucket" {  
  source = "terraform-aws-modules/s3-bucket/aws"  
  version = "3.14.1"  
  # ... other parameters  
}
```

Vision 2026

```
module "s3_bucket" {  
  source = var.s3_module_source  
  version = var.s3_module_version  
  # ... other parameters  
}
```

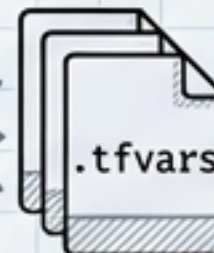
```
variable "s3_module_source" {  
  type = string  
  default = "terraform-aws-modules/s3-bucket/aws"  
  # ...  
}
```

```
variable "s3_module_version" {  
  type = string  
  # ...  
  const = true  
}
```

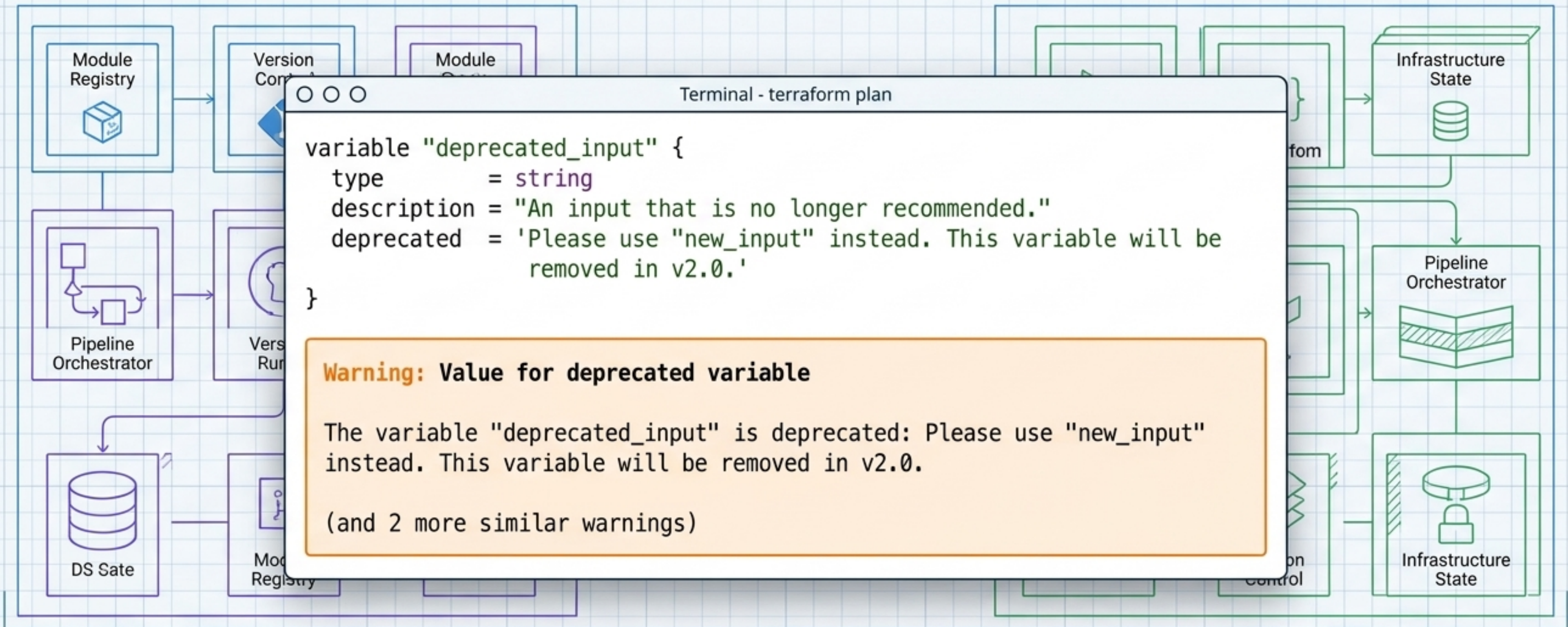
Proving resolution at terraform init. Value is immutable after initialization.



Achieves true environment parity by shifting version control entirely into the .tfvars environment matrix. Zero duplicated logic.

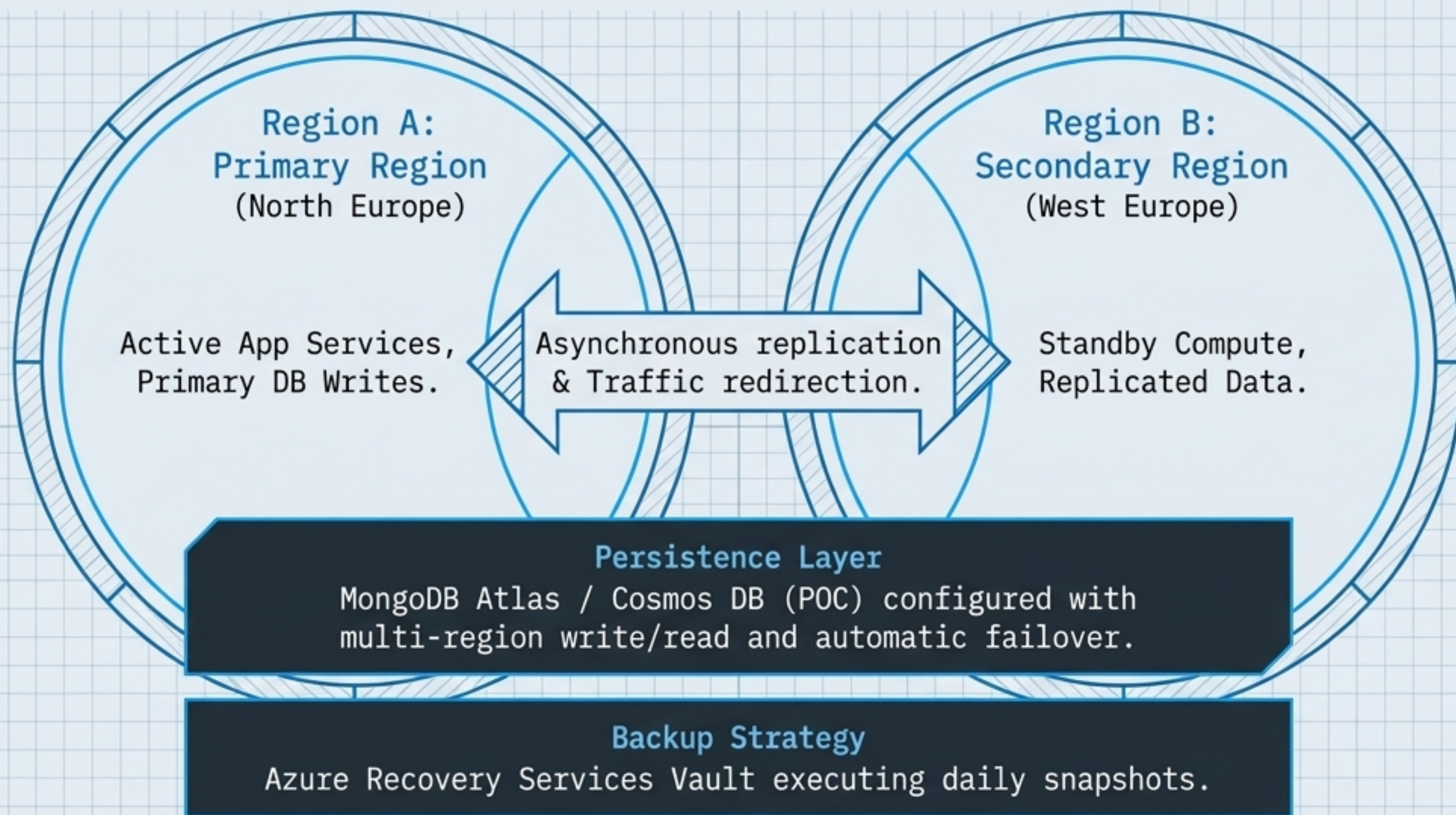


Terraform 1.15: Interface Control & Deprecation



Insight: Platform engineering teams can now safely guide downstream consumers through interface changes without breaking infrastructure pipelines.

BCP & Multi-Region Data Resiliency



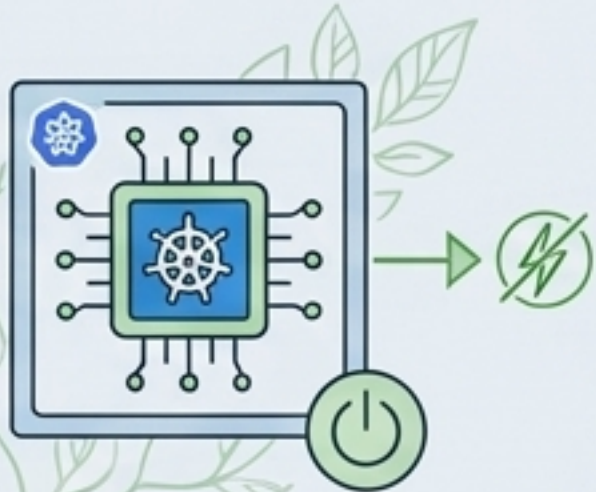
Automated failover sovereignty guarantees data survival and continuity across geographical boundaries.

GreenOps: Carbon-Aware Engineering

AKS Scale-to-Zero

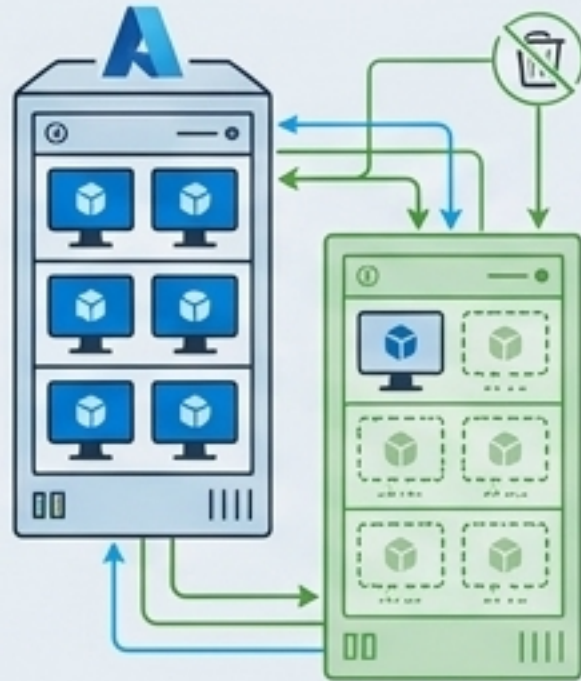
`min_count = 0`

for ML/Jobs pools.
Zero energy consumed
when idle.



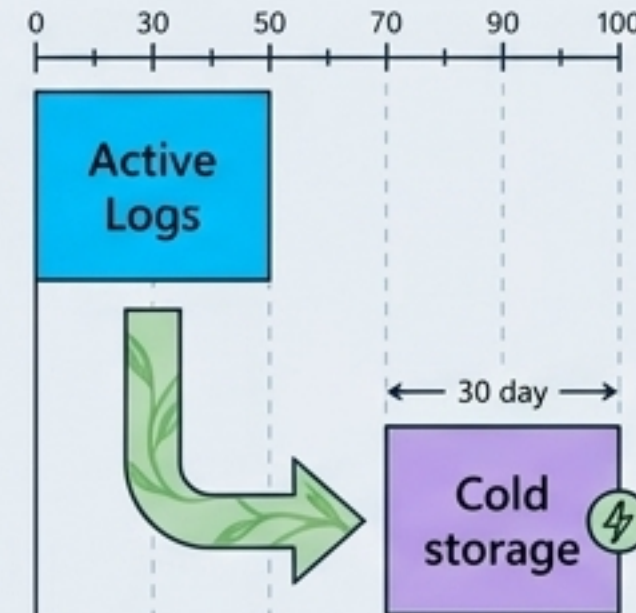
Spot Compute Integration

Azure Spot VMs absorb
non-critical pod workloads,
utilizing waste datacenter
capacity.



Log Retention Governance

Aggressive aging policies
(30 vs 90 days) reduces
Cold Storage energy footprint.



Carbon-Aware Geography

Prioritized deployments
in low-carbon regions
(North Europe - Wind/Hydro).

North Europe - Wind/Hydro



Repository Engineering Metrics & Reality

259

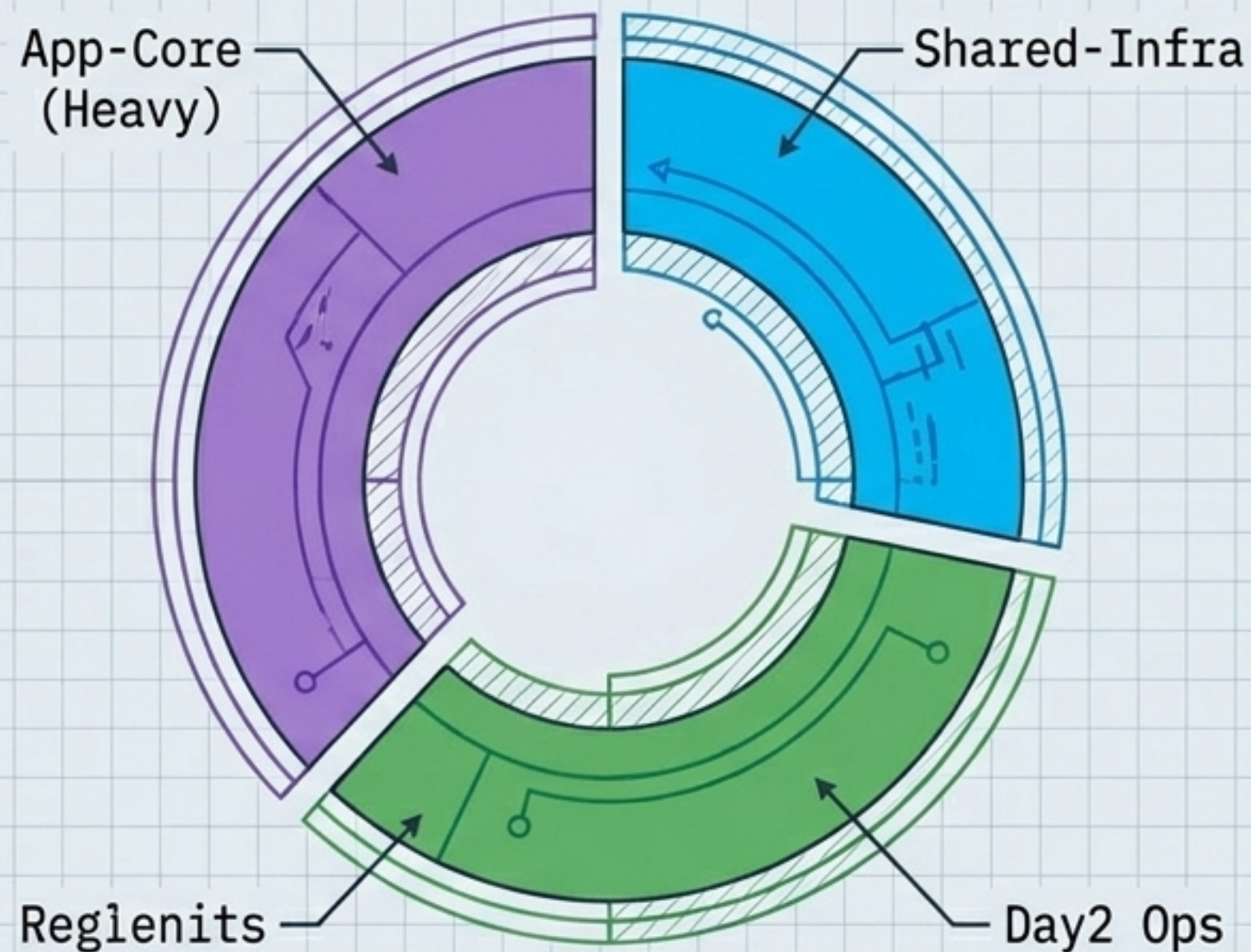
Total Terraform
Resources Managed

170

Orchestrated YAML
Pipelines

515

Configured Files
Across 8 Domains



A unified control plane that treats 100% of the enterprise cloud estate as deterministic, verifiable software.